CPE 324: Digital Signal Processing

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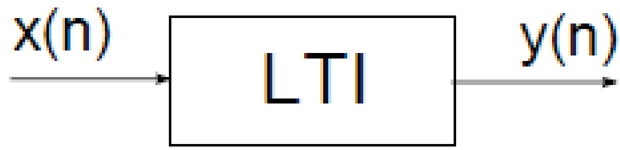
Week 7: Frequency Domain and LTI Systems

- Frequency domain characteristics of LTI systems

- Frequency domain analysis of LTI systems

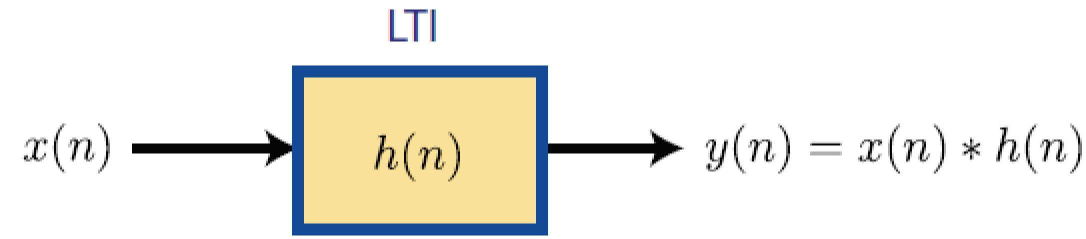
Frequency domain characteristics of LTI systems



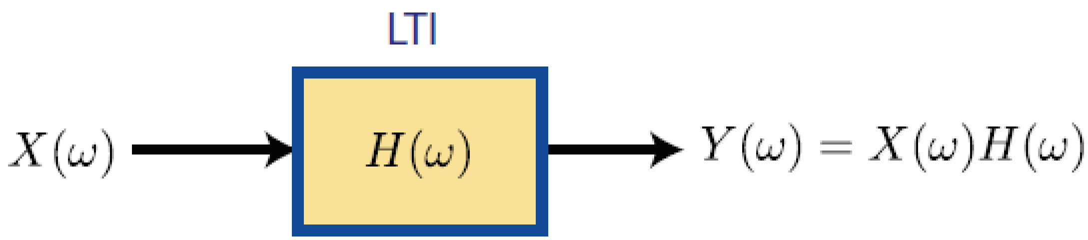
- **LTI:** **Linear Time-Invariant system**
- ***h(n):*** **the impulse response describes the time domain c/s of an LTI systems .**
- ***H(ω):*** **the *frequency response* describes the frequency-domain c/s of an LTI systems .**

$$H(\omega) = \text{DTFT}\,(\,h(n)\,)$$

Frequency domain characteristics of LTI systems



$$h(n) \longleftrightarrow H(\omega)$$



- Recall the response of an LTI system to input signal $x(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} h(n)x(n-k)$$

Frequency domain characteristics of LTI systems

- Excite the system with a complex exponential, i.e. let $x(n) = Ae^{j\omega n}, -\infty < n < \infty, -\pi < \omega < \pi$

$$
\begin{aligned}
y(n) &= \sum_{k=-\infty}^{\infty} h(k)\left[Ae^{j\omega(n-k)}\right] \\
&= A\left[\sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}\right] e^{j\omega n} \\
&= Ae^{j\omega n} H(\omega)
\end{aligned}
$$

where,

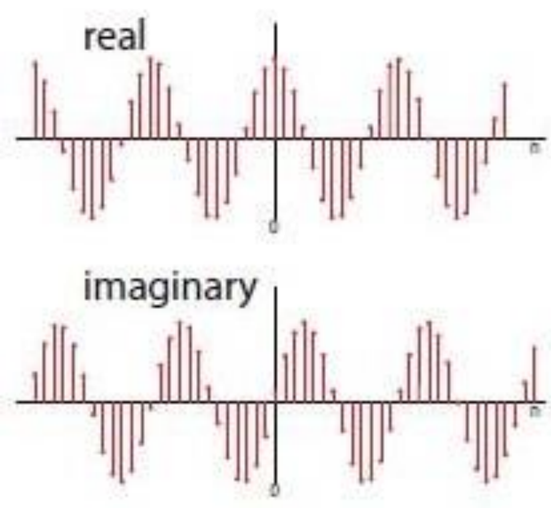
$$
H(\omega) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}
$$

Frequency domain characteristics of LTI systems

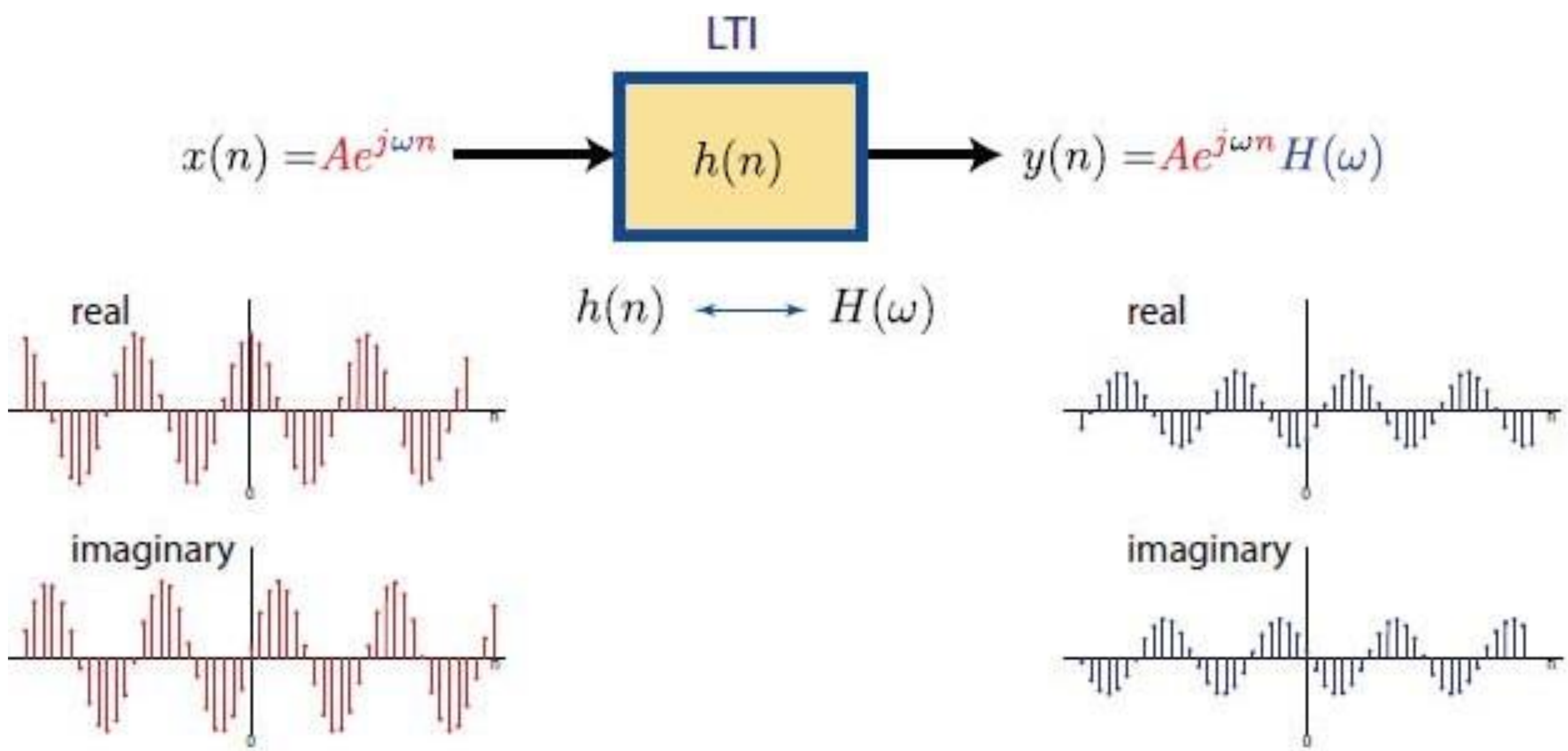
LTI

$$x(n) = Ae^{j\omega n} \longrightarrow \boxed{h(n)} \longrightarrow ???$$

$$h(n) \longleftrightarrow H(\omega)$$



Frequency domain characteristics of LTI systems



Frequency domain characteristics of LTI systems

Observations

- $y(n)$ is in the form of a complex exponential with same frequency as input, multiplied by a factor.
- The complex exponential signal $x(n)$ is called an eigenfunction of the system.
- $H(\omega)$ evaluated at the frequency of the input is the corresponding eigenvalue of the system.

Think of an eigenfunction as a "**special input**" that keeps its shape after passing through the system — only its **size or phase** changes.

The amount it scales by is the **eigenvalue**.

Frequency domain characteristics of LTI systems

$$\begin{aligned} y(n) &= H(\omega)x(n) \\ \text{output} &= \text{scaled input} \\ M \cdot v &= \lambda \cdot v \end{aligned}$$

- Eigenfunction of a system:
  - an input signal that produces an output that differs from the input by a constant (possibly complex) multiplicative factor
  - multiplicative factor is called the eigenvalue

Frequency domain characteristics of LTI systems

$$\underbrace{M \cdot v}_{\text{matrix-vector processing}} = \underbrace{\lambda \cdot v}_{\text{scaled input vector}}$$

$$y(n) = \underbrace{h(n) * Ae^{j\omega n}}_{\text{LTI system processing}} = \underbrace{H(\omega)Ae^{j\omega n}}_{\text{scaled input signal}}$$

- Therefore, a signal of the form $Ae^{j\omega n}$ is an eigenfunction of an LTI system.
- The function $H(\omega)$ represents the associated eigenvalue.

Frequency domain characteristics of LTI systems

- An LTI system can only change the amplitude and phase of a sinusoidal signal. It cannot change the frequency.

- An LTI system with inputs comprised of frequencies from set Ω_0 cannot produce an output signal with frequencies in the set Ω_0^c (i.e., the complement set of Ω_0).
- If you inject a signal comprised of frequencies 1 Hz, 4 Hz and 7Hz into a system and you get an output signal comprised of frequencies 1 Hz and 8 Hz, your system is not LTI.

Frequency domain characteristics of LTI systems

Example: Nonlinear System

Suppose: $x(n) = \cos(2\pi f_1 n + \phi_1) + \cos(2\pi f_2 n + \phi_2)$ is injected into:

$$
\begin{aligned}
y(n) &= x^2(n) \qquad \text{nonlinear system} \\
&= (\cos(2\pi f_1 n + \phi_1) + \cos(2\pi f_2 n + \phi_2))^2 \\
&= \cos^2(2\pi f_1 n + \phi_1) + \cos^2(2\pi f_2 n + \phi_2) + 2\cos(2\pi f_1 n + \phi_1)\cos(2\pi f_2 n + \phi_2) \\
&= \left[\frac{1 + \cos(2\pi(2f_1)n + 2\phi_1)}{2}\right] + \left[\frac{1 + \cos(2\pi(2f_2)n + 2\phi_2)}{2}\right] \\
&\qquad + \left[\frac{\cos(2\pi(f_1 - f_2)n + (\phi_1 - \phi_2)) + \cos(2\pi(f_1 + f_2)n + (\phi_1 + \phi_2))}{2}\right] \\
&= \underbrace{1}_{\text{freq } 0} + \frac{1}{2}[\cos(2\pi(2f_1)n + 2\phi_1) + \cos(2\pi(2f_2)n + 2\phi_2) \\
&\quad + cos(2\pi(f_1 - f_2)n + (\phi_1 - \phi_2)) + \cos(2\pi(f_1 + f_2)n + (\phi_1 + \phi_2))]
\end{aligned}
$$

The output frequencies $(0, 2f_1, 2f_2, f_1 - f_2, f_1 + f_2)$ are different from the input frequencies (f_1, f_2).

Frequency domain characteristics of LTI systems

Observations, cont.

- As $H(\omega)$ is a Fourier transform, it is periodic with period 2π
- $H(\omega)$ is a complex-valued function, can be expressed in polar form

$$H(\omega) = |H(\omega)|e^{\phi\omega}$$

- $h(k)$ is related to $H(\omega)$ through

$$h(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{j\omega k}\,d\omega$$

Frequency domain characteristics of LTI systems

$$H(\omega) = |H(\omega)|e^{j\Theta(\omega)}$$

$$\begin{aligned} |H(\omega)| &\equiv \text{system gain for freq } \omega \\ \angle H(\omega) = \Theta(\omega) &\equiv \text{phase shift for freq } \omega \end{aligned}$$

$$\begin{aligned} y(n) &= H(\omega)Ae^{j\omega n} \\ &= |H(\omega)|e^{j\Theta(\omega)}Ae^{j\omega n} \\ &= A|H(\omega)|e^{j(\omega n+\Theta(\omega))} \end{aligned}$$

Frequency domain characteristics of LTI systems

Real-valued impulse response:

An LTI system with a real-valued impulse response, exhibits symmetry properties

- $|H(\omega)| \Rightarrow$ even function of ω
- $\phi(\omega) \Rightarrow$ odd function of ω.
- Consequently, it is enough to know $|H(\omega)|$ and $\phi(\omega)$ for $0 \leq \omega \leq \pi$

Frequency domain characteristics of LTI systems

Example

- Determine the output sequence of the system with impulse response

$$h(n) = (\frac{1}{2})^n u(n)$$

when the input signal is

$$x(n) = Ae^{j\pi n/2}, -\infty < n < \infty$$

- Solution: evaluate $H(\omega)$

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

$$H(\omega = \frac{\pi}{2}) = \frac{1}{1 + j\frac{1}{2}} = \frac{2}{\sqrt{5}}e^{-j26.6^o}$$

$\omega = \frac{\pi}{2}$ is the frequency of the input signal $x(n)$

Frequency domain characteristics of LTI systems

- Thus, output is

$$y(n) = A\left(\frac{2}{\sqrt{5}}e^{-j26.6^o}\right)e^{j\pi n/2}$$
$$= \frac{2}{\sqrt{5}}Ae^{j(\pi n/2-26.6^o)}, -\infty < n < \infty$$

- system effect on the input: amplitude scale and phase shift
- change frequency, we change amount of scale change and phase shift.

Frequency domain characteristics of LTI systems

Example: Moving Average Filter

- Determine the magnitude and phase of $H(\omega)$ for the three-point moving average (MA) system.

$$y(n) = \frac{1}{3}\left[x(n+1) + x(n) + x(n-1)\right]$$

and plot these functions for $0 \leq \omega \leq \pi$

From given difference equation, we can write

$$h(n) = \frac{1}{3}\left[\delta(n+1) + \delta(n) + \delta(n-1)\right]$$

Frequency domain characteristics of LTI systems

- Solution:

$$h(n) = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$$

consequently,

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \frac{1}{3}(e^{j\omega} + 1 + e^{-j\omega}) = \frac{1}{3}(1 + 2\cos\omega)$$

Hence

$$|H(\omega)| = \frac{1}{3}|1 + 2\cos\omega|$$

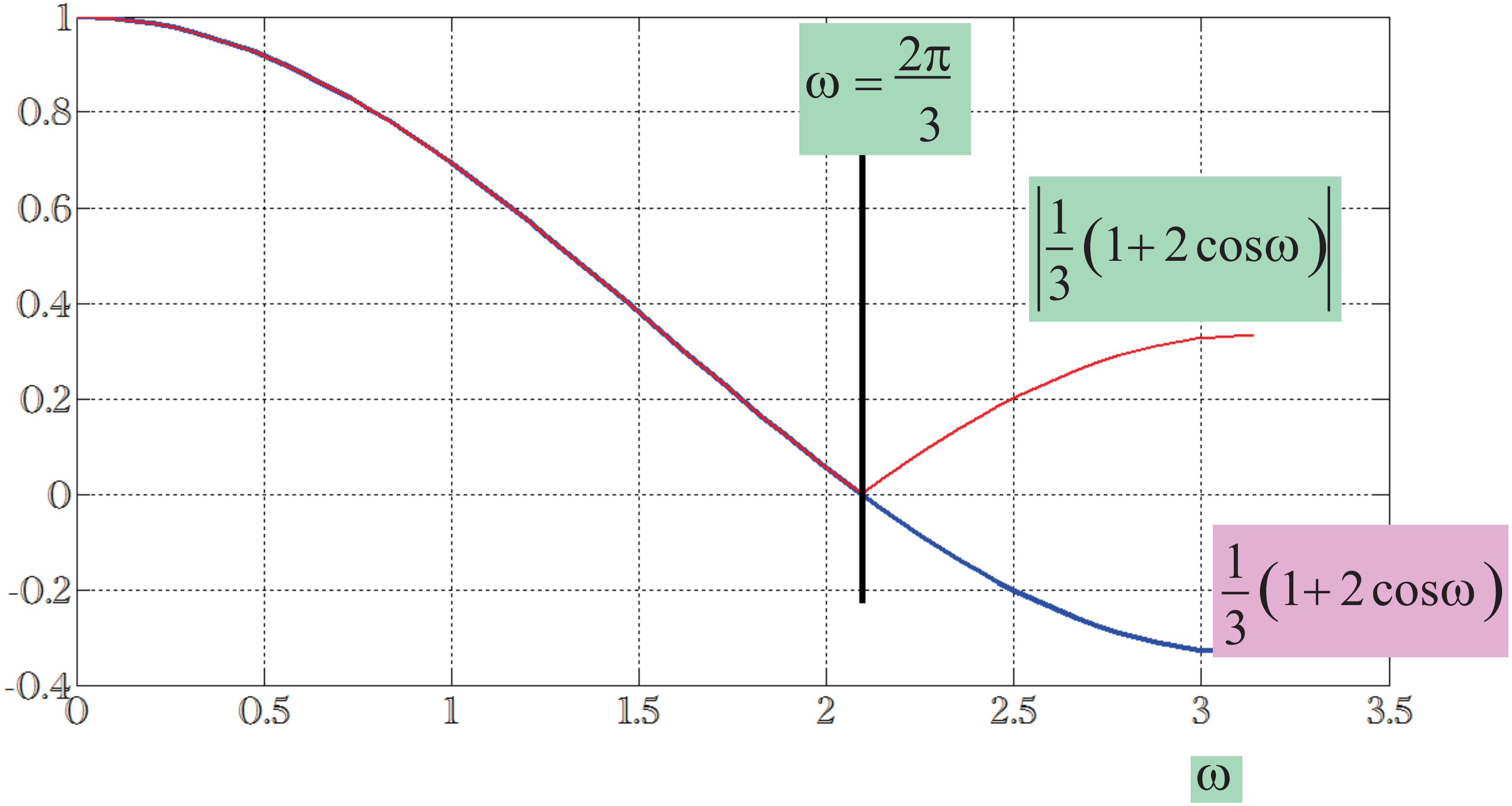
$$\Theta(\omega) = \begin{cases} 0, & 0 \le \omega \le 2\pi/3 \\ \pi, & 2\pi/3 \le \omega < \pi \end{cases}$$

Because $1 + 2\cos\omega$ is a real number. The phase is:

- 0 when $1 + 2\cos\omega > 0$,
- π when $1 + 2\cos\omega < 0$,
- undefined (or jumps) at zeros when it equals 0.

Solve $1 + 2\cos\omega = 0 \to \cos\omega = -\frac{1}{2} \to \omega = \frac{2\pi}{3},\ \frac{4\pi}{3}$. In the interval $[0, \pi)$ the sign-change happens at $\omega = \frac{2\pi}{3}$, which matches the piecewise $\Theta(\omega)$.

Frequency domain characteristics of LTI systems



Frequency domain characteristics of LTI systems

Phase = ?

$$H(\omega) = \frac{1}{3}(1 + 2\cos\omega)$$ → Real Function

$$H(\omega) = |H(\omega)| e^{j\Theta(\omega)}$$ → $$H(\omega) = |H(\omega)| \cos\Theta$$

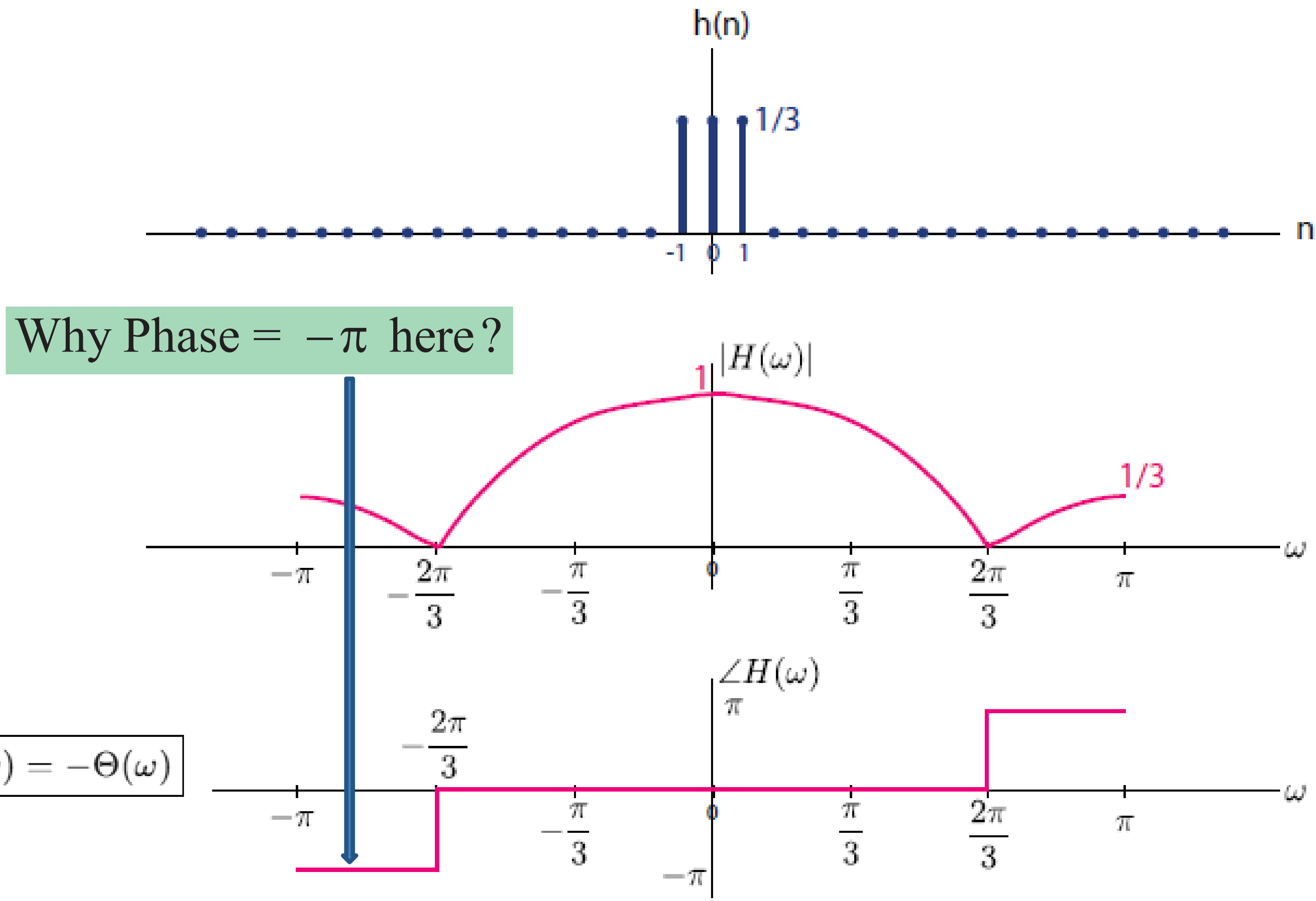
Because imaginary part is zero

$$H(\omega) = |H(\omega)| \cos\Theta$$

$|H(\omega)|$ is always positive

$$\Theta = \begin{cases} 0 & \text{when } H(\omega) \text{ is positive} \\ \pi & \text{when } H(\omega) \text{ is negative} \end{cases}$$

Frequency domain characteristics of LTI systems



$$\Theta(-\omega) = -\Theta(\omega)$$

Frequency domain characteristics of LTI systems

Example: Determine the response of the system

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

for $x(n) = 10 - 5\sin\frac{\pi}{2}n + 20\cos\pi n, -\infty < n < \infty$

- Solution
- Idea Recognise the frequency of each part of the input signal, and find corresponding system response.
- First term 10, fixed signal $\Rightarrow \omega = 0$

$$H(0) = \frac{1}{1 - \frac{1}{2}} = 2$$

Frequency domain characteristics of LTI systems

- $5 \sin \frac{\pi}{2} n$ has a frequency $\omega = \pi/2$, thus

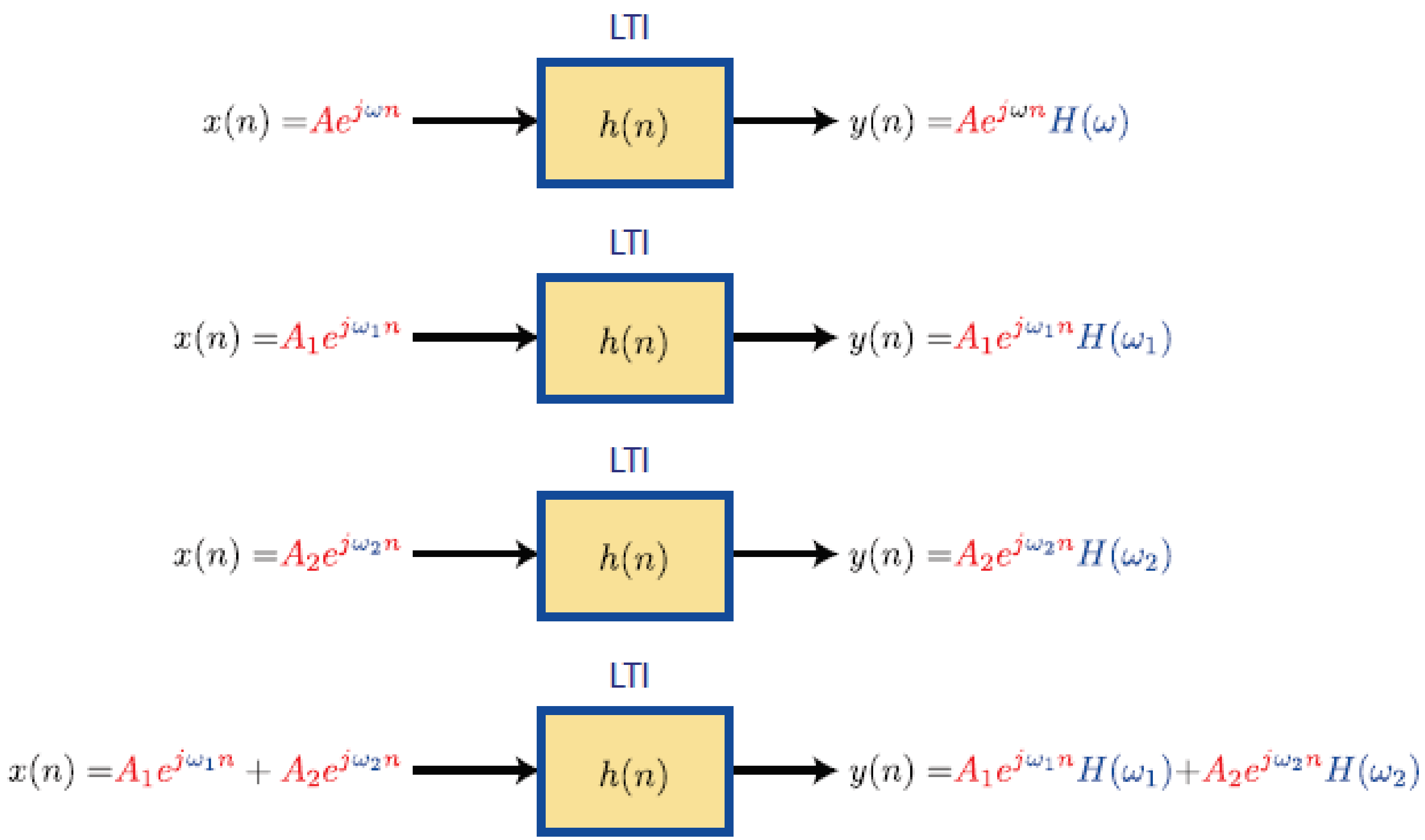
$$H(\frac{\pi}{2}) = \frac{2}{\sqrt{5}} e^{-j26.6^o}$$

- $20 \cos \pi n$ has a frequency $\omega = \pi$

$$H(\pi) = \frac{2}{3}$$

$$\boxed{y(n) = 20 - \frac{10}{\sqrt{5}} \sin\left(\frac{\pi}{2} n - 26.6^\circ\right) + \frac{40}{3} \cos \pi n, \qquad -\infty < n < \infty}$$

Frequency domain characteristics of LTI systems



Frequency domain characteristics of LTI systems

$$x(n) = A_1 e^{j\omega_1 n} + A_2 e^{j\omega_2 n} \longrightarrow \underset{h(n)}{\overset{\text{LTI}}{\boxed{}}} \longrightarrow y(n) = A_1 e^{j\omega_1 n} H(\omega_1) + A_2 e^{j\omega_2 n} H(\omega_2)$$

$$x(n) = \frac{1}{2\pi}\int_{2\pi} X(\omega) e^{j\omega n} d\omega \longrightarrow \underset{h(n)}{\overset{\text{LTI}}{\boxed{}}} \longrightarrow y(n) = \frac{1}{2\pi}\int_{2\pi} X(\omega) H(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi}\int_{2\pi} Y(\omega) e^{j\omega n} d\omega$$

- An LTI system changes the amplitudes and phase shifts of the individual frequency components within $x(n)$ to produce $y(n)$.
- $H(\omega)$ dictates how frequency ω is changed in the signal.

Basics of filtering operation

Frequency domain characteristics of LTI systems

The General Case

- Most general case: input to the system is an arbitrary linear combination of sinusoids of the form

Input phase

$$x(n) = \sum_{i=1}^{L} A_i \cos(\omega_i n + \phi_i), -\infty < n < \infty$$

Where $\{A_i\}$ and $\{\phi_i\}$ amplitude and phase of corresponding sinusoidal component i .

- System response will be of the form:

output phase

$$y(n) = \sum_{i=1}^{L} A_i |H(\omega_i)| \cos\left[\omega_i n + \phi_i + \Theta(\omega_i)\right]$$

Where $|H(\omega_i)|$ and $\Theta(\omega_i)$ are the magnitude and phase imparted by the system to the individual frequency components of the input signal.

Frequency domain characteristics of LTI systems

Periodic Input Signals

- Let the input signal $x(n)$ to a stable LTI be periodic with fundamental period N.

- Using the Fourier series representation of a periodic signal

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}, \quad k = 0, 0, \ldots, N-1$$

- Evaluating the system response for each complex exponential

$$H\left(\frac{2\pi k}{N}\right) = H(\omega)|_{\omega=2\pi kn/N}, \quad k = 0, 1, \ldots, N-1$$

Frequency domain characteristics of LTI systems

$$y(n) = \sum_{k=0}^{N-1} c_k H\left(\frac{2\pi k}{N}\right) e^{j2\pi kn/N}, \quad -\infty < n < \infty$$

- LTI system response to a periodic input signal is also periodic with the same period N, with coefficients related by

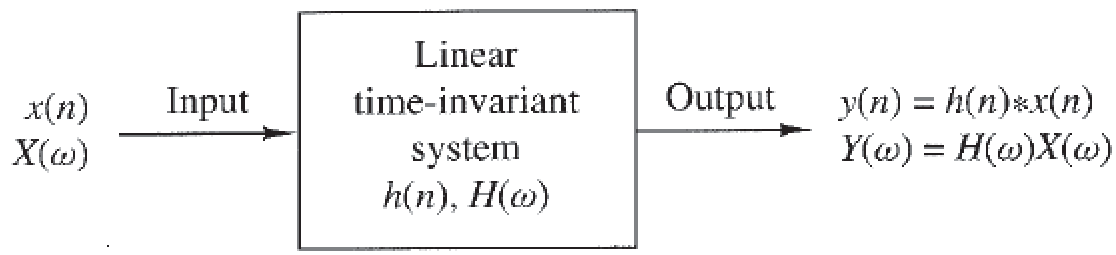
$$d_k = c_k H\left(\frac{2\pi k}{N}\right), \quad k = 0, 1, ..., N-1$$

FS coefficients of output signal

FS coefficients of input signal

Frequency domain characteristics of LTI systems

Aperiodic Input Signals



- Let $\{x(n)\}$ ne the aperiodic input sequence, $\{y(n)\}$ output sequence, and $\{h(n)\}$ unit sample response.
- By Convolution theorem

  $Y(\omega) = H(\omega)X(\omega)$

  In polar form, magnitude and phase of the output signal:

  $|Y(\omega)| = |H(\omega)||X(\omega)|$

  $\angle Y(\omega) = \angle H(\omega) + \angle X(\omega)$

- Output of an LTI system can NOT contain frequency components that are not contained in the input signal.

Frequency domain characteristics of LTI systems

- Energy density spectra of input and output

$$|Y(\omega)|^2 = |H(\omega)|^2|X(\omega)|^2$$
$$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$$

- Energy of the output signal

$$E_y = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{yy}(\omega)d\omega$$
$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} |H(\omega)|^2 S_{xx}(\omega)d\omega$$

Frequency domain characteristics of LTI systems

Example:

- A linear time-invariant system is characterized by its impulse response

$$h(n) = \left(\frac{1}{2}\right)^2 u(n)$$

- Determine the spectrum and energy density spectrum of the output signal when the system is excited by the signal

$$x(n) = \left(\frac{1}{4}\right)^2 u(n)$$

Frequency domain characteristics of LTI systems

- Frequency response of the system

$$H(\omega) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^2 e^{-j\omega n}$$
$$= \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

- Similiarly, Fourier transform of the input sequence

$$X(\omega) = \frac{1}{1 - \frac{1}{4}e^{-j\omega}}$$

- Spectrum of the output signal

$$Y(\omega) = H(\omega)X(\omega)$$
$$= \frac{1}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})}$$

Frequency domain characteristics of LTI systems

- energy density spectrum

$$S_{xx}(\omega) = |Y(\omega)|^2 = |H(\omega)|^2 |X(\omega)|^2$$
$$= \frac{1}{(\frac{5}{4} - \cos\omega)(\frac{17}{16} - \frac{1}{2}\cos\omega)}$$

Frequency domain characteristics of LTI systems

Frequency response from system function:

$$H(\omega) = H(z)|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

System function evaluated at unit circle given that unit circle is contained in ROC.

Mostly, we are interested in $|H(\omega)|^2$ which for real $h(n)$ is given by:

$$|H(\omega)|^2 = H(\omega)H^*(\omega) = H(\omega)H(-\omega) = H(z)H(z^{-1})|_{z=e^{j\omega}}$$

Frequency domain characteristics of LTI systems

Example: Determine $|H(\omega)|^2$ for the system

$$y(n) = -0.1y(n-1) + 0.2y(n-2) + x(n) + x(n-1)$$

The system function is

$$H(z) = \frac{1+z^{-1}}{1+0.1z^{-1}-0.2z^{-2}}$$

ROC is $|z| > 0.5$. Hence $H(\omega)$ exists.

$$H(z)H(z^{-1}) = \frac{1+z^{-1}}{1+0.1z^{-1}-0.2z^{-2}} \cdot \frac{1+z}{1+0.1z-0.2z^2}$$

$$= \frac{2+z+z^{-1}}{1.05+0.08(z+z^{-1})-0.2(z^{-2}+z^{-2})}$$

Frequency domain characteristics of LTI systems

By evaluating $H(z)H(z^{-1})$ on the unit circle, we obtain

$$|H(\omega)|^2 = \frac{2 + 2\cos\omega}{1.05 + 0.16\cos\omega - 0.4\cos 2\omega}$$

Frequency domain analysis of LTI systems

LTI Systems as Frequency-Selective Filters

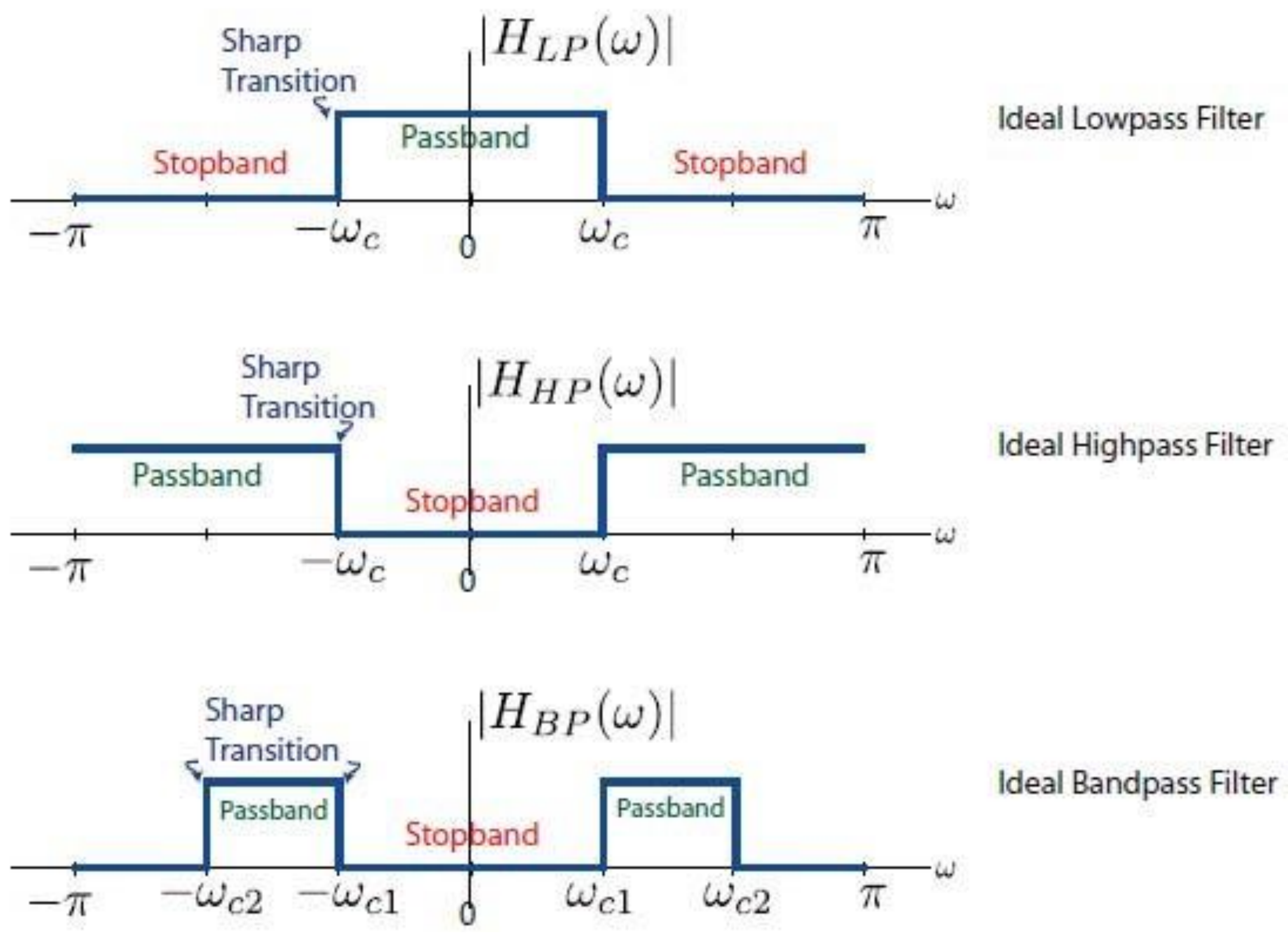
- Filter: device that discriminates, according to some attribute of the input, what passes through it

- For LTI systems, given $Y(\omega) = X(\omega)H(\omega)$
  - $H(\omega)$ acts as a kind of weighting function or spectral shaping function of the different frequency components of the signal

$$\text{LTI system} \Longleftrightarrow \text{Filter}$$

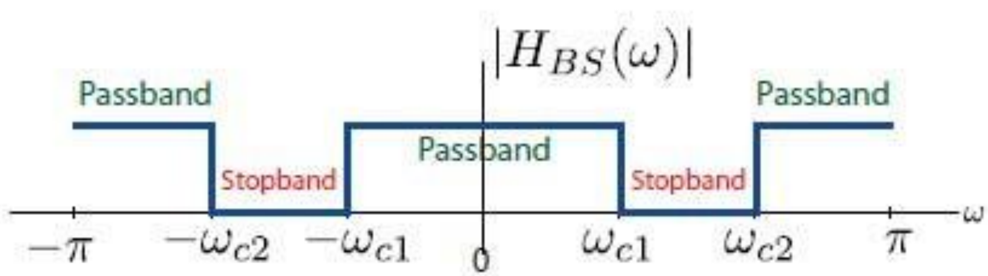
Frequency domain analysis of LTI systems

Ideal Filters

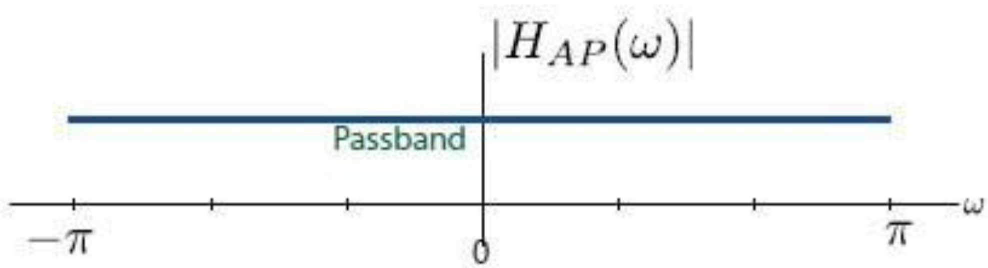


Frequency domain analysis of LTI systems

Ideal Filters



Ideal Badstop Filter



Ideal All-pass Filter

Frequency domain analysis of LTI systems

What we saw previously that:

- LTI system changes the amplitude of input
- Delays (introduces the phase shift) it

$$H(\omega) = |H(\omega)|e^{j\Theta(\omega)}$$

$$\begin{aligned} |H(\omega)| &\equiv \text{system gain for freq } \omega \\ \angle H(\omega) = \Theta(\omega) &\equiv \text{phase shift for freq } \omega \end{aligned}$$

$$\begin{aligned} y(n) &= H(\omega)Ae^{j\omega n} \\ &= |H(\omega)|e^{j\Theta(\omega)}Ae^{j\omega n} \\ &= A|H(\omega)|e^{j(\omega n+\Theta(\omega))} \end{aligned}$$

Frequency domain analysis of LTI systems

Ideal Filters

- Common characteristics:
  - flat (typically unity for $C = 1$) frequency response magnitude in passband and zero frequency response in stopband
  - linear phase; for constants C and n_0

$$H(\omega) = \begin{cases} Ce^{-j\omega n_0} & \omega_1 < |\omega| < \omega_2 \\ 0 & \text{otherwise} \end{cases}$$

If the phase is **linear with frequency**, then **all frequency components** of a signal experience the **same time delay** when passing through the system.

- A signal is made up of sinusoids at different frequencies.
- If each sinusoid gets delayed **by the same amount**, the signal's shape stays the same.
- That happens only if phase = linear in frequency.

Frequency domain analysis of LTI systems

Ideal Filters

- Suppose $H(\omega) = Ce^{-j\omega n_0}$ for all ω:

$$\delta(n) \overset{\mathcal{F}}{\longleftrightarrow} 1$$

$$\delta(n - n_0) \overset{\mathcal{F}}{\longleftrightarrow} 1 \cdot e^{-j\omega n_0}$$

$$C \cdot \delta(n - n_0) \overset{\mathcal{F}}{\longleftrightarrow} C \cdot 1 \cdot e^{-j\omega n_0} = Ce^{-j\omega n_0}$$

Output is simply scaled and delayed version of input

- Therefore, $h(n) = C\delta(n - n_0)$ and:

$$y(n) = x(n) * h(n) = x(n) * C\delta(n - n_0) = Cx(n - n_0)$$

Frequency domain analysis of LTI systems

Ideal Filters

- Therefore for ideal linear phase filters:

$$H(\omega) = \begin{cases} Ce^{-j\omega n_0} & \omega_1 < |\omega| < \omega_2 \\ 0 & \text{otherwise} \end{cases}$$

Signal becomes zero

  - signal components in stopband are annihilated
  - signal components in passband are shifted (and scaled by passband gain which is unity (for $C = 1$))

Frequency domain analysis of LTI systems

Phase versus Magnitude

Phase very important in Filter design !

What's more important?

We will study in detail filter characteristics after S-2 exam !

Frequency domain analysis of LTI systems

Invertibility of LTI Systems

Why Invert?

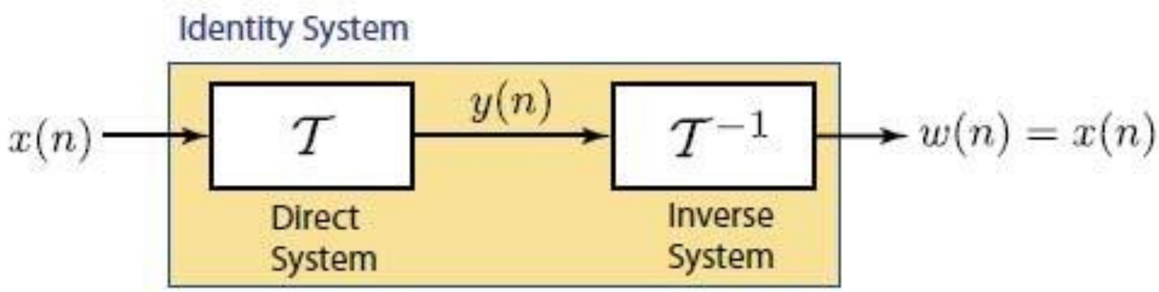
- There is a fundamental necessity in engineering applications to undo the unwanted processing of a signal.
  - reverse intersymbol interference in data symbols in telecommunications applications to improve error rate; called equalization
  - correct blurring effects in biomedical imaging applications for more accurate diagnosis; called restoration/enhancement
  - increase signal resolution in reflection seismology for improved geologic interpretation; called deconvolution

Frequency domain analysis of LTI systems

Invertibility of Systems

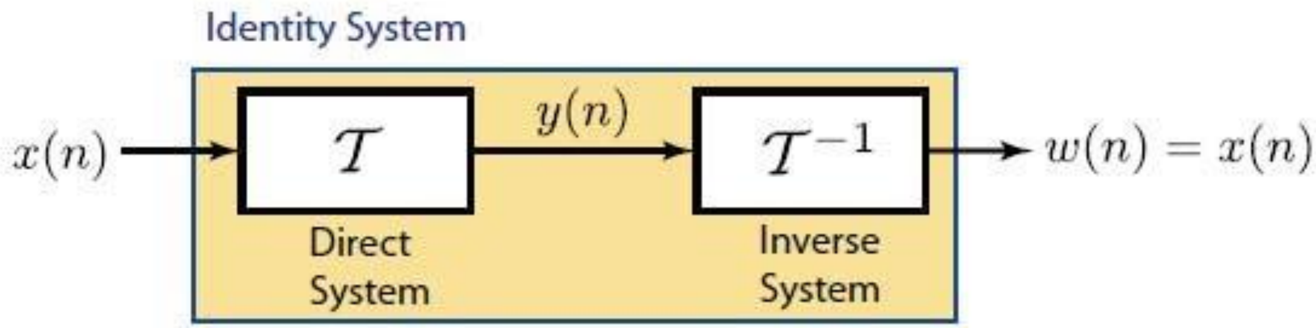
- Invertible system: there is a one-to-one correspondence between its input and output signals
- the one-to-one nature allows the process of reversing the transformation between input and output; suppose

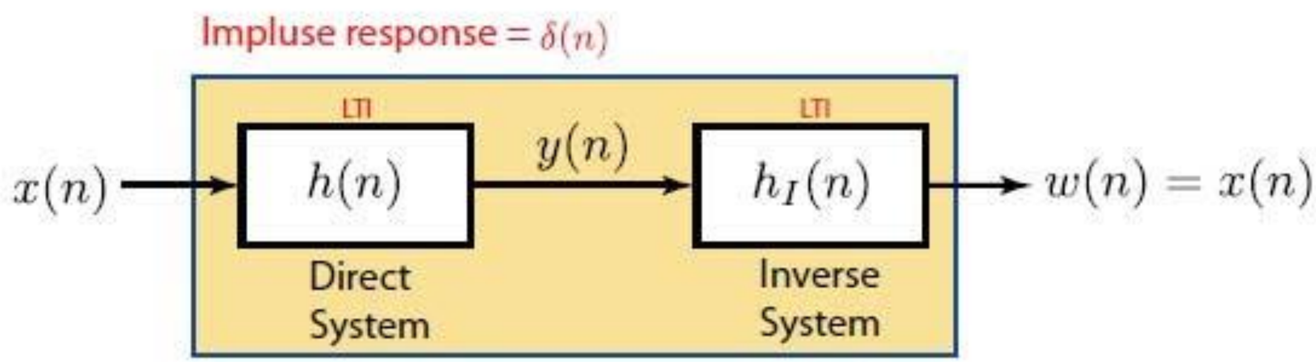
$$\begin{aligned} y(n) &= \mathcal{T}[x(n)] \quad \text{where } \mathcal{T} \text{ is one-to-one} \\ w(n) &= \mathcal{T}^{-1}[y(n)] = \mathcal{T}^{-1}[\mathcal{T}[x(n)]] = x(n) \end{aligned}$$



Frequency domain analysis of LTI systems

Invertibility of LTI Systems





Frequency domain analysis of LTI systems

Invertibility of LTI Systems

- Therefore,

$$h(n) * h_I(n) = \delta(n)$$

- For a given $h(n)$, how do we find $h_I(n)$?
- Consider the $z-$domain

$$\begin{aligned} H(z)H_I(z) &= 1 \\ H_I(z) &= \frac{1}{H(z)} \end{aligned}$$

Frequency domain analysis of LTI systems

Invertibility of Rational LTI Systems

- Suppose, $H(z)$ is rational:

$$
\begin{aligned}
H(z) &= \frac{A(z)}{B(z)} \\
H_I(z) &= \frac{B(z)}{A(z)} \\
\text{poles of } H(z) &= \text{zeros of } H_I(z) \\
\text{zeros of } H(z) &= \text{poles of } H_I(z)
\end{aligned}
$$

Frequency domain analysis of LTI systems

Example

Determine the inverse system of the system with impulse response $h(n) = (\frac{1}{2})^n u(n)$.

$$a^n u(n) \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z| > |a|$$

- $H(z) = \frac{1}{1-\frac{1}{2}z^{-1}}$, ROC: $|z| > \frac{1}{2}$.
- Therefore,

$$H_I(z) = \frac{1}{H(z)} = 1 - \frac{1}{2}z^{-1}$$

- By inspection,

$$h_I(n) = \delta(n) - \frac{1}{2}\delta(n-1)$$

Frequency domain analysis of LTI systems

Another Example

Determine the inverse system of the system with impulse response $h(n) = \delta(n) - \frac{1}{2}\delta(n-1)$.

$$
\begin{aligned}
H(z) &= \sum_{n=-\infty}^{\infty} h(n) z^{-n} = \sum_{n=-\infty}^{\infty} \left[\delta(n) - \frac{1}{2}\delta(n-1)\right] z^{-n} \\
&= 1 - \frac{1}{2} z^{-1} \\
H_I(z) &= \frac{1}{1 - \frac{1}{2} z^{-1}}
\end{aligned}
$$

But in this case two choices for inverse system !

Frequency domain analysis of LTI systems

$a^n u(n)$ $\longleftrightarrow$ $\frac{1}{1-az^{-1}}$ $|z| > |a|$

$-a^n u(-n-1)$ $\longleftrightarrow$ $\frac{1}{1-az^{-1}}$ $|z| < |a|$

There are two possibilities for inverses:

- Causal + stable inverse ($|z| > \frac{1}{2}$ includes unit circle):

$$h_I(n) = \left(\frac{1}{2}\right)^n u(n)$$

- Anticausal + unstable inverse ($|z| < \frac{1}{2}$ does not include unit circle):

$$h_I(n) = -\left(\frac{1}{2}\right)^n u(-n-1)$$

The End